


Enumerating Weak Order Unit Fubini Rankings

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Plan for Talk

- Definitions
  - Tuples and Weak Order
  - Fubini Rankings
  - Unit Interval Parking Functions
  - Unit Fubini Rankings
- Motivation
- Main Result (Enumerating Sets)




Definitions: Tuples and Weak Order

Tuples and Weak Order




Definition (Tuple)

A *tuple* of length n is a finite set of ordered elements, denoted by $\alpha = (a_1, a_2, \ldots, a_n) \in [n]^n$.

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Definition (Weakly decreasing rearrangement)

The *weakly decreasing rearrangement* of α is $\alpha^{\downarrow} = (a'_1, a'_2, \ldots, a'_n)$ such that $a'_i \geq a'_{i+1}$ for all $i \in [n-1]$.




Definitions: Fubini Rankings

Fubini Rankings

Definition (Fubini ranking)

Let $\alpha = (a_1, a_2, ..., a_n) \in [n]^n$. Then, α is a *Fubini ranking* of length n if for all $r \in [n]$, if k entries of α are equal to r, then the next larger value in α is $r + k$. The set of Fubini rankings of length n is denoted FR_n.

Fubini Rankings



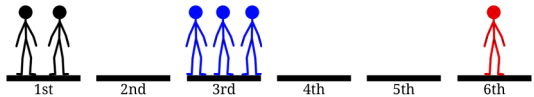

Example

Take $\alpha = (\underline{3}, \underline{1}, \underline{3}, \underline{6}, \underline{3}, \underline{1}) \in \mathrm{FR}_6$, and read as "competitor i placed in a_i-th rank."

Fubini Rankings

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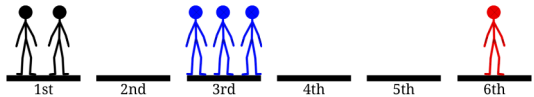
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Note: FR_n is permutation invariant!

Lemma [1, Ager-Hart et al., 2026]

Given $\alpha = (a_1, a_2, \ldots, a_n) \in \mathrm{FR}_n$, let $\alpha^{\uparrow} = (a'_1, a'_2, \ldots, a'_n)$ be its weakly increasing rearrangement. Then, for all $i \in [n]$, either $a'_i = i$ or $a'_i = a'_{i-1}$.

Fubini Rankings

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Example

- Take $\alpha = (3, 1, 1) \in \mathrm{FR}_3$. We have $\alpha^{\uparrow} = (a'_1, a'_2, a'_3) = (1, 1, 3)$.

Fubini Rankings

Lemma [1, Ager-Hart et al., 2026]

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Example

- Take $\alpha = (3, 1, 1) \in \mathrm{FR}_3$. We have $\alpha^{\uparrow} = (a'_1, a'_2, a'_3) = (1, 1, 3)$.
- Take $\alpha = (2, 1, 1) \notin \mathrm{FR}_3$. We have $\alpha^{\uparrow} = (a'_1, a'_2, a'_3) = (1, 1, 2)$.




Definitions: Unit Interval Parking Functions

Unit Interval Parking Functions




Definition (Unit interval parking function)

Let $\alpha = (a_1, a_2, ..., a_n) \in [n]^n$, and read as "car i has parking preference a_i." Then, α is a *unit interval parking function* of length n if for all $i \in [n]$, car i can park in its preferred spot a_i or the spot right after, which is $a_i + 1$, i.e., all cars park and no parking spots are left empty. The set of unit interval parking functions of length n is denoted UPF_n.

Unit Interval Parking Functions





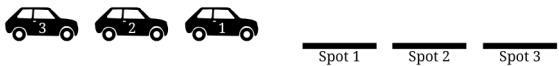
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Take $\alpha = (1, 1, 2) \in \mathrm{UPF}_3$.

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Take $\alpha = (1, 1, 2) \in \mathrm{UPF}_3$. Let's park!



Unit Interval Parking Functions





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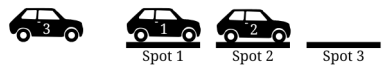
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Caution: UPF_n is not permutation invariant!

Unit Interval Parking Functions




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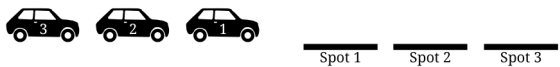
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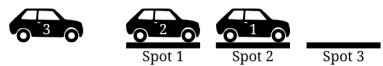
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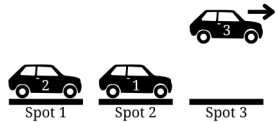
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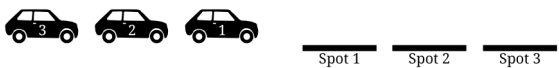
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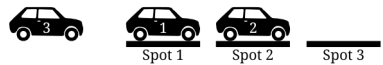


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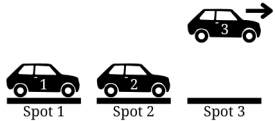


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Definitions: Unit Fubini Rankings

Unit Fubini Rankings





Lemma (Characterization of unit Fubini rankings)

For all $n \geq 1$, $\alpha = (a_1, a_2, \ldots, a_n) \in [n]^n$ is in $\mathrm{FR}_n \cap \mathrm{UPF}_n$ if and only if the weakly increasing rearrangement of α, denoted by $\alpha^{\uparrow} = (a'_1, a'_2, \ldots, a'_n)$, satisfies the following:

(1) either $a'_i = i$ or $a'_i = a'_{i-1}$ for all $i \in [n]$

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(1) Required by Fubini rankings [1, Ager-Hart et al., 2026].

(2) Required by unit interval parking functions (Lemma).

Motivation

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Theorem [3, Elder et al., 2025]

For all $n \geq 1$,

$$|\mathrm{FR}_n^{\uparrow} \cap \mathrm{UPF}_n^{\uparrow}| = F_{n+1}$$

where F_{n+1} is the $(n+1)$th Fibonacci number. The terms for $1 \leq n \leq 7$ are:

$$1, 2, 3, 5, 8, 13, 21.$$

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1. For $n = 1$, $|\mathrm{FR}_1^{\uparrow} \cap \mathrm{UPF}_1^{\uparrow}| = 1$.
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2. For $n = 2$, $|\mathrm{FR}_2^{\uparrow} \cap \mathrm{UPF}_2^{\uparrow}| = 2$.
3. The cases are disjoint and exhaustive, satisfying a recursion relation.

Motivation





Question: What is the cardinality of

- $\mathrm{UPF}_n^{\downarrow}$
- $\mathrm{FR}_n^{\downarrow} \cap \mathrm{UPF}_n^{\downarrow}$?




Main Result (Enumerating Sets)

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First, we want $|\mathrm{FR}_n^{\uparrow} \cap \mathrm{UPF}_n^{\uparrow}| = |\mathrm{FR}_n^{\downarrow} \cap \mathrm{UPF}_n^{\downarrow}|$.

Proof of Main Result





Lemma ($\mathrm{FR}_n \cap \mathrm{UPF}_n$ Permutation Invariance)

For all $n \geq 1$, the set $\mathrm{FR}_n \cap \mathrm{UPF}_n$ is permutation invariant.

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Proof by Example

Take $\alpha = (4, 5, 2, 1, 2, 5) \in \mathrm{FR}_6 \cap \mathrm{UPF}_6$.

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- Cars not in a box park in their preferred spot.
- First car in each box parks in its preferred spot.
- Second car in the same box parks in spot right after its preferred spot.
- Entries in a box are indistinguishable and can be permuted.

Proof of Main Result





Definition (rev(α))

Let $\alpha = (a_1, a_2, \ldots, a_n)$. Define rev($\alpha$) to reverse the order of the entries in α such that rev(α) $= (a_n, \ldots, a_2, a_1)$.

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Corollary

For all $n \geq 1$, rev($\alpha^{\uparrow}$) $= \alpha^{\downarrow}$ and rev($\alpha^{\downarrow}$) $= \alpha^{\uparrow}$.

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Definition (rev(α))

Let $\alpha = (a_1, a_2, \ldots, a_n)$. Define $\mathrm{rev}(\alpha)$ to reverse the order of the entries in α such that $\mathrm{rev}(\alpha) = (a_n, \ldots, a_2, a_1)$.

Corollary

For all $n \geq 1$, $\mathrm{rev}(\alpha^{\uparrow}) = \alpha^{\downarrow}$ and $\mathrm{rev}(\alpha^{\downarrow}) = \alpha^{\uparrow}$.

Lemma

For all $n \geq 1$, $\alpha \in \mathrm{FR}_n^{\downarrow} \cap \mathrm{UPF}_n^{\downarrow}$ if and only if $\mathrm{rev}(\alpha) \in \mathrm{FR}_n^{\uparrow} \cap \mathrm{UPF}_n^{\uparrow}$.

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For all $n \geq 1$, $\text{rev}(\alpha^{\uparrow}) = \alpha^{\downarrow}$ and $\text{rev}(\alpha^{\downarrow}) = \alpha^{\uparrow}$.

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For all $n \geq 1$, $\alpha \in \text{FR}_n^{\downarrow} \cap \text{UPF}_n^{\downarrow}$ if and only if $\text{rev}(\alpha) \in \text{FR}_n^{\uparrow} \cap \text{UPF}_n^{\uparrow}$.

Thus, $|\text{FR}_n^{\uparrow} \cap \text{UPF}_n^{\uparrow}| = |\text{FR}_n^{\downarrow} \cap \text{UPF}_n^{\downarrow}|$.

Proof of Main Result





Theorem

For all $n \geq 1$,

$$|\mathrm{FR}_n^{\uparrow} \cap \mathrm{UPF}_n^{\uparrow}| = |\mathrm{FR}_n^{\downarrow} \cap \mathrm{UPF}_n^{\downarrow}| = |\mathrm{UPF}_n^{\downarrow}| = F_{n+1}$$

where F_{n+1} is the $(n+1)$th Fibonacci number. The terms for $1 \leq n \leq 7$ are:

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Now, we want $|\mathrm{FR}_n^{\uparrow} \cap \mathrm{UPF}_n^{\uparrow}| = |\mathrm{FR}_n^{\downarrow} \cap \mathrm{UPF}_n^{\downarrow}| = |\mathrm{UPF}_n^{\downarrow}|$.

Proof of Main Result




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- We already saw $(2, 1, 1) \notin \mathrm{FR}_3^{\downarrow} \cup \mathrm{UPF}_3^{\downarrow}$.

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- Consider $(3, 3, 1) \notin \mathrm{FR}_3^{\downarrow}$. Car 2 fails to park, and $\alpha \notin \mathrm{UPF}_3^{\downarrow}$.

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- Consider $(3,3,1) \notin \mathrm{FR}_3^{\downarrow}$. Car 2 fails to park, and $\alpha \notin \mathrm{UPF}_3^{\downarrow}$.
- Consider $(3,1,1) \in \mathrm{FR}_3^{\downarrow}$. All cars park, so $\alpha \in \mathrm{UPF}_3^{\downarrow}$.



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where F_{n+1} is the $(n+1)$th Fibonacci number. The terms for $1 \leq n \leq 7$ are:

$$1, 2, 3, 5, 8, 13, 21.$$

We already know the cardinality is F_{n+1} [3, Elder et al., 2025].

Proof of Main Result





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Question: While recursion is nice, can we find a closed formula?

Yes!

Proof of F_{n+1} Enumeration





Lemma

For all $n \geq 1$, if $\alpha = (a_1, a_2, \ldots, a_n)$, then $k \leq \lfloor \frac{n}{2} \rfloor$ entries $i \in \alpha$ can repeat once.

Proof of F_{n+1} Enumeration





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For $\alpha = (a_1, a_2, \ldots, a_n) \in \mathrm{FR}_n \cup \mathrm{UPF}_n$, if entries $i, j \in \alpha$ are repeats, they are not consecutive numbers.

Example

Consider $(1, 1, 2, 2) \notin \mathrm{FR}_4 \cup \mathrm{UPF}_4$ due to invalid entries and parking conflicts.

Proof of F_{n+1} Enumeration




Proof

Given $\alpha = (a_1, a_2, \ldots, a_n) \in \mathrm{FR}_n \cap \mathrm{UPF}_n$, choose $k \leq \lfloor \frac{n}{2} \rfloor$ distinct entries $i \in \alpha$ to repeat from the $n - k$ options (subtract the numbers greater than and consecutive to each i).

Proof of F_{n+1} Enumeration





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$$\sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n-k}{k}.$$

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Given $\alpha = (a_1, a_2, \ldots, a_n) \in \mathrm{FR}_n \cap \mathrm{UPF}_n$, choose $k \leq \lfloor \frac{n}{2} \rfloor$ distinct entries $i \in \alpha$ to repeat from the $n - k$ options (subtract the numbers greater than and consecutive to each i). Counting the variations, we have

$$\sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n-k}{k}.$$

For $1 \leq n \leq 7$, the summation yields

$$1, 2, 3, 5, 8, 13, 21,$$

which are the $(n+1)$th Fibonacci numbers, F_{n+1}.

Example of F_{n+1} Enumeration





Example

Determine $|\mathrm{FR}_4^{\downarrow} \cap \mathrm{UPF}_4^{\downarrow}| = |\mathrm{UPF}_4^{\downarrow}|$.

Example of F_{n+1} Enumeration



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Determine $|\mathrm{FR}_4^{\downarrow} \cap \mathrm{UPF}_4^{\downarrow}| = |\mathrm{UPF}_4^{\downarrow}|$. We have

$$\sum_{k=0}^{\lfloor \frac{4}{2} \rfloor} \binom{4-k}{k} = \underbrace{\binom{4-0}{0}}_{1} + \underbrace{\binom{4-1}{1}}_{3} + \underbrace{\binom{4-2}{2}}_{1} = 5.$$

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$$\mathrm{FR}_4^{\downarrow} \cap \mathrm{UPF}_4^{\downarrow} = \mathrm{UPF}_4^{\downarrow} \begin{cases} (4,3,2,1) & (k = 0 \text{ repeats}) \\ (4,3,1,1) & (k = 1 \text{ repeat}) \\ (4,2,2,1) & (k = 1 \text{ repeat}) \\ (3,3,2,1) & (k = 1 \text{ repeat}) \\ (3,3,1,1) & (k = 2 \text{ repeats}). \end{cases}$$

Summary




- Definitions
  - Tuples and Weak Order
  - Fubini Rankings
  - Unit interval parking functions
  - Unit Fubini Rankings
- Motivation
- Main Result (Sets Enumerated by F_{n+1})

Future Work

- Given $\alpha \in \mathrm{FR}_n$, park the cars. If a car's preferred spot is occupied, the car will continue down the street until it finds an empty spot to park.
- Given $\alpha \in \mathrm{UPF}_n$, follow the parking rules stated in the definition.

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Definition (Total Displacement)

Let $\alpha \in \mathrm{FR}_n \cup \mathrm{UPF}_n$. The total number of spots each car parked away from its preferred spot is called the *total displacement* of α, denoted $d(\alpha)$.

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Let $\alpha \in \mathrm{UPF}_n$. Then $0 \leq d(\alpha) \leq n-1$.

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Let $\alpha \in \mathrm{UPF}_n$. Then $0 \leq d(\alpha) \leq n-1$.

Proposition (Total displacement of $\mathrm{FR}_n \cap \mathrm{UPF}_n$)

Let $\alpha \in \mathrm{FR}_n \cap \mathrm{UPF}_n$. Then $0 \leq d(\alpha) \leq \lfloor \frac{n}{2} \rfloor$.

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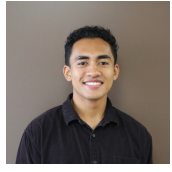
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